Today: - hitting probability continued:

- first of two patterns in an i.i.d. sequence

ex: 0 0 1 0 1 1 0 0 1 1 0 ...

first occurrence of either

001

or

011

(which happens first)

- hitting time: $T_A := \inf\{n : X_n \in A\}$.

formula for $g(x) := E_x[T_A]$

(a BVP determines g).

examples:

- Gambler's ruin with $p = 1/2$:

$$E_x\left[T_{\{0,N\}}\right] = x(N-x)$$

if $x = N/2$ this $= N^2/4$.

- birth & death chains.
- renewal chains.

- Martingales: (X_n) is a martingale if

$$\forall n, \quad E[X_{n+1} \mid X_0, \dots, X_n] = X_n.$$

Optional stopping theorem:

Let τ be a stopping time for martingale (X_n). If either

(i) $\tau < \infty$ and $\sup_n |X_n| \leq C < \infty$ for some C a.s.

or

(ii) $\sup_{n < \tau} |X_{n+1} - X_n| \leq C < \infty$ a.s. & $E[\tau] < \infty$

then

$$E[X_\tau] = E[X_0].$$

(fair game is still fair if you quit at a stopping time).

usage: Gambler's ruin with $p = 1/2$.

(X_n) is a martingale since

$$E[X_{n+1} \mid X_0, \dots, X_n, \ 0 < X_n < N]$$

$$= (X_n - 1)\cdot \tfrac{1}{2} + (X_n + 1)\cdot \tfrac{1}{2}$$

$$= X_n$$

To use optional stopping with

$$\tau = \inf\{n : X_n \in \{0, N\}\}$$

$$E[X_\tau] = E[X_0]$$

($|X_n| \le N$ for this MC).

if $X_0 = x$, then $E[X_0] = x$ and

$$E[X_\tau] = N \cdot P_x(X_\tau = N) + 0 \cdot P_x(X_\tau = 0)$$

$$= N h(x)$$

so $h(x) = x/N$.

Pattern probability

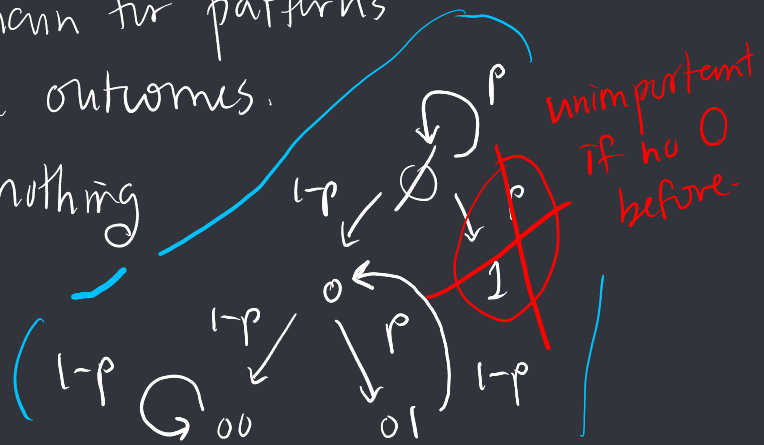
in i.i.d. Bernoulli(p) sequence

find $P(001 \text{ occurs before } 011)$.

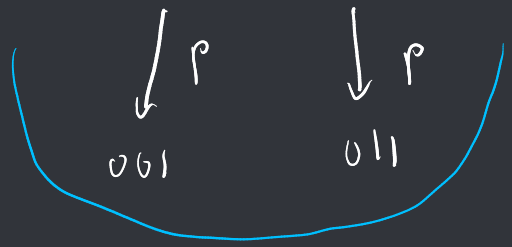
build Markov chain for patterns leading to these outcomes.

Initially there is nothing

then, 0 or 1



Markov chain of partial patterns relevant to 001 & 011.



Desired probability is $P_\emptyset(X_T = 001)$

where $T = \inf\{n : X_n \in \{001, 011\}\}$.

Let $h(x) = P_x(X_T = 001)$ & compute h:

$h(001) = 1, \qquad h(011) = 0$

&

$$h(\phi) = p\,h(\phi) + (1-p)\,h(0)$$

$$h(0) = (1-p)\,h(00) + p\,h(01)$$

$$\vdots \quad = (1-p) + p(1-p)\,h(0)$$

$$h(\phi) = h(0)$$

$$h(00) = (1-p)\,h(00) + p\,\underbrace{h(001)}_{=1}$$

$$p\,h(00) = p \quad \therefore \quad h(00) = 1.$$

$$h(01) = (1-p)\,h(0)$$

exercise: find typo.

$$p\,h(0) = p(1-p)$$

$$h(0) = 1-p$$

$$h(\phi) = 1-p$$

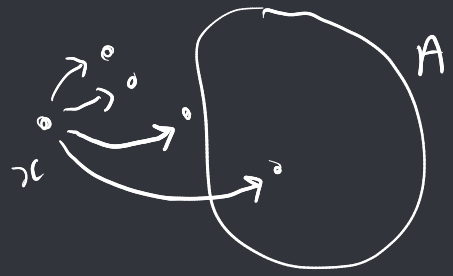
Hitting time

fix $A \subset S$. Define $T_A = \min\{n : X_n \in A\}$.

Let $g(x) = E_x[T_A]$.

Then g is characterized by the BVP:

$$g(x) = \begin{cases} 0 & x \in A \\ 1 + \sum_y p(x,y) g(y) & x \notin A \end{cases}$$



take one step, after which remaining expected time

$$= \sum_y p(x,y)\,g(y)$$

Gambler's ruin with $p = 1/2$: let $A = \{0, N\}$.

Then

$$g(0) = g(N) = 0$$

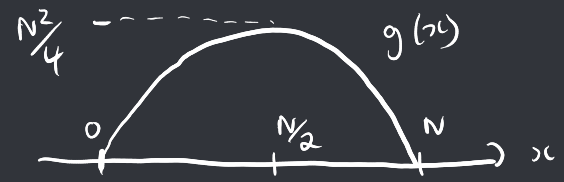
and for $0 < x < N$,

$$g(x) = 1 + \tfrac{1}{2}\left(g(x+1) + g(x-1)\right)$$

Can check $x(N-x)$ is a solution:

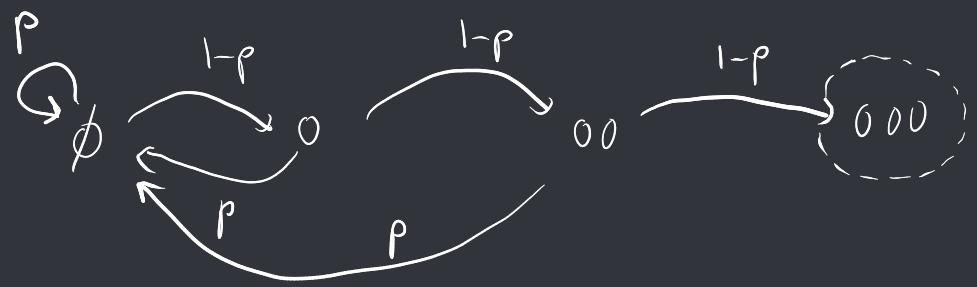
$$1 + \frac{1}{2}\Big(\underbrace{(x+1)(N-(x+1))}_{\substack{x(N-x) \\ +(-x) \\ +N \\ -(x+1)}} + \underbrace{(x-1)(N-(x-1))}_{\substack{x(N-x) \\ +x \\ -N \\ +(x-1)}}\Big)$$

$$1 + \frac{1}{2}\Big(2x(N-x) - 2\Big) = x(N-x)$$



ex: hitting time of pattern

find expected time to see 000 in i.i.d. Bernoulli(p) sequence.



$g(x) = E_x[T]$; $T = \inf\{n : X_n = 000\}$.

$g(000) = 0$ and $g(\emptyset) = 1 + p\,g(\emptyset) + (1-p)\,g(0)$

$$g(0) = 1 + p\,g(\emptyset) + (1-p)\,g(00)$$

$$g(00) = 1 + p\,g(\emptyset) + 0$$

$$g(0) = 1 + p\,g(\emptyset) + (1-p)(1 + p\,g(\emptyset))$$

$$g(\emptyset) = \frac{1}{1-p} + g(0)$$

$$= \frac{1}{1-p} + 1 + 1 - p + g(\emptyset)(p + p(1-p))$$

$$g(\emptyset)(1 - p(2-p)) = \frac{1}{1-p} + 1 + 1 - p$$

for $p = \frac{1}{2}$, $\quad g(\emptyset)\left(1 - \frac{1}{2}\cdot\frac{3}{2}\right) = 2 + 1 + \frac{1}{2}$

$$g(\emptyset) = 4(7/2) = 14.$$

<u>exercise</u>: generalize formula to $\underbrace{000\cdots0}_{n \text{ in a row.}}$

(note: answer should be $\leq n2^n$).

Martingale approach to hitting time for G.R. w/ $p = 1/2$:

Claim: $X_n^2 - n$ is a martingale (ignoring boundaries: i.e. if $p(x, x+1) = p(x, x-1) = 1/2$).

How is this used? Optional stopping:

$$E_x[X_\tau^2 - \tau] = E_x[X_0^2 - 0] = x^2$$

$$\therefore \quad E[\tau] = \underbrace{E_x[X_\tau^2]}_{} - x^2$$

$$= N^2 P_x(X_\tau = N)$$

$$= N^2 \cdot x/N = Nx$$

$$\therefore \quad E[\tau] = Nx - x^2 = x(N-x)$$

Let's check $X_n^2 - n$ is a m.g:

$$E[X_{n+1}^2 - (n+1) \mid X_0, \ldots, X_n]$$

$$= \frac{1}{2}\cdot\left((x_n+1)^2 + (x_n-1)^2\right) - (n+1)$$

$$= \frac{1}{2}\left(x_n^2+2x_n+1+x_n^2-2x_n+1\right) - (n+1)$$

$$= x_n^2+1-(n+1)$$

$$= x_n^2-n.$$

h-transform.

hitting probab. of A before B, starting from x

Given disjoint $A, B \subset S$ & $h(x)$ as above.

① - $h(\underbrace{X_{n \wedge T}}_{\min(n,T)})$ is a martingale

$\min(n,T)$; $T = T_{A \cup B}$ (hitting time of $A \cup B$).

② - $q(x,y) := p(x,y) \, h(y) / h(x)$ defines transition probability & has

$$P(X_{n+1} = y \mid X_n = x, \; X_T = A, \; T > n) = q(x,y)$$

q is the h-transformed trans. mtx.
" " h-transform of p.

① explain: if $T > n$ then

$$E[h(X_{n+1}) \mid X_0, \dots, X_n]$$

$$= \sum_y p(X_n, y)\, h(y)$$

$$= h(X_n) \quad \checkmark$$

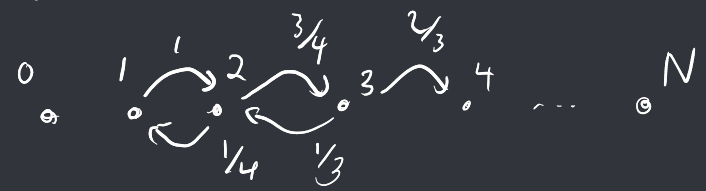
② can check using Bayes' formula

ex: G.R. w/ $p = 1/2$

Know $h(x) = x/N$ so

$$q(x, x+1) = p(x, x+1)\,\frac{h(x+1)}{h(x)}$$

$$= \frac{1}{2}\,\frac{x+1/N}{x/N} = \frac{1}{2}\left(1 + \frac{1}{x}\right)$$

sim.

$$q(x, x-1) = \frac{1}{2}\left(1 - \frac{1}{x}\right)$$